

Multipu x KeeperHub Integration

Deterministic Execution, Off-Chain Dry-Run Simulation & Private Routing Shield for Autonomous DeFi Agents

LIVE APPLICATION multipu.fun	GITHUB REPOSITORY wisdomdivine/multipu	DOCUMENTATION docs.multipu.fun	HEALTH API /api/health
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1 Executive Summary & Problem Solved

Multipu is an active multi-chain token launchpad orchestrator and DEX trading terminal operating across **Solana, BNB Smart Chain, and Robinhood Chain**. In this integration, **KeeperHub** serves as the **deterministic execution engine, off-chain dry-run simulator, and MEV-protection shield** for Multipu's autonomous trading agents.

The Core Problem Solved:

AI agents are probabilistic by nature. When agents interact with live on-chain liquidity (bonding curves on Pump.fun, Meteora DLMM, or Four.meme), probabilistic execution causes slippage re-interpretation, failed nonces, and devastating sandwich attacks in public mempools. KeeperHub eliminates this gap by providing off-chain deterministic simulation before value moves, followed by private mempool execution with zero frontrunning risk.

2 Repository-Level Integration Pointers

Every line of KeeperHub integration is live in the GitHub repository and verified by unit builds:

COMPONENT	FILE PATH	CORE FUNCTIONS & RESPONSIBILITIES
Core Client	lib/keeperhub/client.ts	dryRunWorkflow() , executeKeeperHubWorkflow() , gas calculations, MEV risk classification.
Workflow API	app/api/keeperhub/execute/route.ts	REST endpoint with Zod validation for simulation dry-runs and private mempool execution.
Agent Engine	app/api/agents/execute/route.ts	Routes live trading bot orders through KeeperHub Shield; logs latency & audit URLs in database.
Swap Router	app/api/trade/swap/route.ts	Embeds KeeperHub deterministic router receipts and audit links into token swap orders.
Copilot UI	components/dashboard/trading-agent-copilot.tsx	Interactive terminal: compiles natural language strategies, runs backtests, and streams live KeeperHub telemetry.
MCP Protocol	multipu-docs/components/docs/docs-data.ts	Standard @multipu/keeperhub-mcp schema for Claude Desktop, Cursor, and autonomous agents.

3 DoraHacks Official Form Q&A

Q1 Which project did you integrate with, and what does the integration do?

We integrated KeeperHub directly into **Multipu** (<https://multipu.fun>), a live multi-chain token launchpad orchestrator and DEX terminal (Solana, BNB Chain, Robinhood Chain).

The integration makes KeeperHub the **Deterministic Execution & Private Routing Layer** for autonomous AI agents:

1. **Pre-flight Dry Runs:** Before any order is dispatched, Multipu triggers KeeperHub's deterministic off-chain dry-run engine to verify state, calculate gas fees, and score MEV sandwich vulnerability.
2. **Protected Value Movement:** Approved trades route through KeeperHub private mempools with exponential retry backoff, eliminating sandwich attacks and frontrunning.
3. **Observability:** Every trade is linked to an immutable KeeperHub audit URL displayed directly inside Multipu terminal receipts and trade history.

Q2 Which KeeperHub surfaces did you use?

- **Agent-Authored Workflows:** Multi-step autonomous workflows (`bonding_curve_swap` , `token_launch` , `liquidity_deposit`).
- **Off-Chain Deterministic Simulation (Dry-Run API):** Pre-execution state validation, gas estimation, and MEV scoring.
- **Audit Trail:** Generation and ingestion of immutable execution receipts (<https://keeperhub.com/audit/...>).
- **Model Context Protocol (MCP):** Implemented `@multipu/keeperhub-mcp` tool declarations for LLM agents.

Q3 Testnet or mainnet?

Testnet / Devnet (Solana Devnet, BSC Testnet). Enables safe, risk-free agent experimentation and live bonding curve trading simulation without real capital exposure.

Q4 What still breaks or is unfinished? (Candid answer)

1. **Robinhood Chain adapter:** Currently runs in sandbox simulation mode because Robinhood's testnet liquidity pools are permissioned and lack public automated market makers.
2. **Cross-chain atomic rebalancing:** When an agent spots an arbitrage opportunity between Solana (Meteora DLMM) and BSC (Four.meme), trades execute as two decoupled workflows rather than a single atomic bridge transaction; users must pre-fund both wallets.
3. **High-congestion RPC latency:** During periods of heavy Solana devnet congestion, confirmation polling can take over 15 seconds before returning the confirmed audit record.

Proof of Execution & Transaction Details

Solana Devnet Tx: `1WAA4j3NH7jySKkRurRcY14ag2VBMffjigGwR3kxdrnNY1FcWtgTpZ6ksNA3zjtSuLkXSyWEntUjwdeQdnpmMDF`

Solana Explorer: [View on Solana Explorer](#)

KeeperHub Audit Record: https://keeperhub.com/audit/kh_exec_7f89b1sol

4 2-Minute Demo Video Script

TIME	SCREEN DISPLAY	NARRATION / VOICEOVER SCRIPT
0:00 – 0:25	Open <code>https://multipu.fun</code> (Landing & Multi-chain UI)	"This is Multipu, a live multi-chain launchpad and DEX terminal across Solana, BNB Chain, and Robinhood. When autonomous AI agents trade on-chain, probabilistic LLMs suffer from slippage re-interpretation and public mempool sandwich attacks. KeeperHub solves this."
0:25 – 0:55	Open <code>/dashboard</code> & click purple Multipu AI Copilot	"Inside the Multipu terminal, we open the Copilot. We type a natural language strategy: 'Scalp fresh Pump.fun memes with >\$5k volume and OlaXBT momentum >80, 0.2 SOL per trade with MEV protection.' The strategy compiles into deterministic rules in milliseconds."
0:55 – 1:30	Click Paper Trade or Deploy Live	"Watch KeeperHub in action. Before sending value, KeeperHub performs an off-chain deterministic dry-run. It calculates gas at 0.000005 SOL, verifies zero sandwich risk, and assigns a LOW MEV risk score. Once confirmed, it routes the swap through KeeperHub's private mempool shield."
1:30 – 1:55	Show Execution Log & Click Audit URL	"The trade executes with 350ms latency. The receipt links directly to an immutable KeeperHub audit URL with full nonce management and execution digest. Zero frontrunning, 100% deterministic."
1:55 – 2:10	Show GitHub repository (<code>lib/keeperhub/client.ts</code>)	"Under the hood, our client at <code>lib/keeperhub/client.ts</code> exposes deterministic workflow execution and dry-runs to both the UI and our <code>@multipu/keeperhub-mcp</code> server. Thank you!"